

# Joshua Veitch-Michaelis

📍 Gainesville, FL

## General Information

---

**Languages:** English (Native), German (A1), Spanish (A2-B1)

**Pronouns:** He/Him

## Education

---

- |              |                                                                                                                                                                                                                                                     |          |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>PhD</b>   | <b>Mullard Space Science Laboratory, University College London</b> , Space and Climate Physics <ul style="list-style-type: none"><li>Fusion of LIDAR with stereo camera data — an assessment</li><li>Supervised by Prof. Jan Peter Muller</li></ul> | Jan 2017 |
| <b>MPhys</b> | <b>University of Warwick</b> , Physics <ul style="list-style-type: none"><li>First class honours</li></ul>                                                                                                                                          | Jan 2012 |

## Experience

---

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>University of Chicago</b> , South Pole Telescope Winterover <ul style="list-style-type: none"><li>Operate and maintain the South Pole Telescope, on-site support for Event Horizon Telescope observations, critical hardware/software support.</li></ul>                                                                                                                                                                                                                                                   | South Pole, Antarctica<br>Jan 2022 – Jan 2024<br>1 year 1 month  |
| <b>ETH Zurich</b> , Postdoctoral Researcher in AI for remote sensing <ul style="list-style-type: none"><li>Joint with Restor Eco</li><li>Developing machine learning algorithms for ecosystem conservation, based on remotely sensed data. Developed Restor's tree crown delineation pipeline.</li></ul>                                                                                                                                                                                                      | Switzerland<br>Jan 2022 – Jan 2023<br>1 year 1 month             |
| <b>Wisconsin-Madison University</b> , IceCube Winterover Experiments Operator <ul style="list-style-type: none"><li>Operate and maintain the IceCube Neutrino Observatory at the South Pole, providing critical hardware and software support (on-call 24/7 year-round)</li></ul>                                                                                                                                                                                                                             | South Pole, Antarctica<br>Jan 2020 – Jan 2022<br>2 years 1 month |
| <b>Liverpool John Moores University</b> , Postdoctoral Researcher in Drone Image Analysis <ul style="list-style-type: none"><li>Led instrument design for a real-time, drone-mounted endangered species detection system using thermal/visible imaging and deep learning.</li><li>Collaborations with Save The Elephants (Kenya), WWF, ZSL (London), Durrell (Madagascar) and Knowsley Safari.</li><li>Development of an automatic spectral calibration pipeline for the 2m Liverpool Telescope</li></ul>     | Jan 2018 – Jan 2020<br>2 years 1 month                           |
| <b>Frontier Development Lab (Europe)</b> , Researcher <ul style="list-style-type: none"><li>Worked on a team developing a flood segmentation algorithm for deployment on a satellite, supported by UNICEF and the European Space Agency (ESA). Selected from over 400 applicants.</li><li>Research presented orally at Humanitarian Aid and Disaster Response workshop at NeurIPS 2019 (hadr.ai).</li><li>Launched code on a D-Orbit satellite in June 2021 and it has been flight-tested in space.</li></ul> | Jan 2019 – Jan 2019<br>1 month                                   |
| <b>I3D Robotics Ltd</b> , Computer Vision/Robotics Engineer <ul style="list-style-type: none"><li>Designed bespoke 3D imaging systems; stereo, LIDAR, laser triangulation, high-speed image processing tools (&gt; 400 fps).</li><li>Led a successful InnovateUK-funded project for smart agriculture; an embedded system for deep learning-based fruit detection in greenhouses.</li><li>Clients included robotics, heavy industry (steel), agriculture and nuclear.</li></ul>                               | Jan 2018 – Jan 2018<br>1 month                                   |

**IS Instruments Ltd**, STFC CASE PhD Student; Photonics Engineer

Jan 2013 – Jan 2017

- Led end-to-end design of a commercial IoT miniature spectrometer. Obtained UK government grant funding, PCB design and embedded firmware (Atmel AVR), opto-mechanical design and end-user software development.

4 years 1 month

## Honors and Awards

---

**2021, 2024:** Antarctica Service Medal (Wintered Over), National Science Foundation (USA)

**2019:** Google Cloud Research Credit LJMU (\$5k) to support research into RGB-Thermal data fusion for ecology

**2017:** Remote Sensing and Photogrammetry Society (RSPSoc) award for best PhD thesis

**2016:** SIM Scholar, Worshipful Company of Scientific Instrument Makers. Postgraduate award for excellence in instrumentation research, nominated by UCL.